Construction Intelligence (CI-DF) as a Decision Science:

A Spatial-Temporal Adaptive Framework for Engineering Project Contracting

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Abstract

This study introduces SAITCO / Construction Intelligence Decision Framework (CI-DF) as an integrated construction intelligence ecosystem that transforms engineering project contracting from fragmented, document-based, experience-driven practice into an adaptive, decision-centric, self-learning digital intelligence system operating across the full project lifecycle. The paper argues that persistent problems of cost overrun, payment delay, weak qualification, reactive claims, and construction disputes are not isolated failures; they are symptoms of a missing decision architecture across the project lifecycle. The proposed framework establishes Construction Decision Theory (CDT), where each project is treated as a dynamic system governed by data, constraints, objectives, and uncertainty. CI-DF integrates SCIM, CDS, PLDS, GACM, SECIM, DCIM, and CPOM. A central contribution is the principle that system validity does not mean producing the same price or decision everywhere; rather, validity means producing contextually correct outputs for each location, time, currency, market, resource environment, and risk profile. The framework is conceptually validated through five applied cases: a financial dispute and residual claim, a risk-adjusted pricing and contractual safeguard case, a project delivery and claims-recovery portfolio case, an organizational structuring and qualification readiness case, and a cost intelligence / BOQ analysis case. The cross-case analysis demonstrates that construction disputes are delayed manifestations of earlier decision failures and that cost is not the final output of construction intelligence, but one of its decision inputs.

This paper introduces a fundamental principle: construction system validity is defined by contextual correctness rather than uniform outputs across different environments.

AI Use Disclosure:

Language editing, formatting, and document structuring assistance were used; the conceptual framework, case studies, and professional data are provided by the author.

Keywords

Construction Intelligence; Decision Systems; Construction Management; Project Lifecycle; Risk Pricing; Claims Management; Spatial-Temporal Applicability; Cost Intelligence; Saudi Arabia; Digital Construction.

1. Introduction

The construction sector remains one of the most important drivers of national development, infrastructure growth, public investment, industrial expansion and private-sector productivity. Yet many construction organizations still manage projects as a sequence of fragmented professional activities: pricing is separated from qualification, qualification is separated from contractual risk, execution is separated from financial control, and claims are treated as legal events after damage has already accumulated.

This paper proposes a different view. Engineering project contracting should be understood as a decision system. Each project is not merely a set of drawings, quantities, activities, payments and correspondence; it is a lifecycle of decisions made under constraints and uncertainty.

SAITCO / CI-DF is therefore defined as an integrated Construction Intelligence ecosystem that transforms engineering project contracting from fragmented, document-based, experience-driven practice into an adaptive, decision-centric, self-learning digital intelligence system operating across the full project lifecycle.

2. Problem Domain: Fragmented Construction Decision Systems

Problem = Fragmented Practice + Static Pricing + Weak Qualification + Reactive Decisions

The problem is not only inaccurate prices. The deeper problem is systemic fragmentation across the full contracting lifecycle. Current practice often produces separate decisions for pricing, qualification, contracting, execution, cash flow, claims and closeout. However, a construction project behaves as an integrated decision environment, where failure in one layer can produce delayed damage in another layer.

3. Transformation Paradigm and Objective Layer

Construction -> Intelligent Decision Science

Objective = Transform(Contracting, from Document-Based Practice, to Adaptive Digital Intelligence)

The fundamental idea is not to create a pricing program only. The objective is to transform construction contracting from a document-based professional practice into a digital intelligent decision science. Under CI-DF, construction reads data, analyzes uncertainty, produces decisions, measures outcomes, learns from results, and adapts to the market, place and time.

4. Construction Decision Theory (CDT)

Project = Decision System

Decision = f(Data, Constraints, Objectives, Uncertainty)

Construction Decision Theory is the theoretical foundation of CI-DF. It assumes that the basic unit of construction analysis is not the BOQ item alone, the contract clause alone or the schedule activity alone. The basic unit is the construction decision in its context.

5. Construction Intelligence (CI) as a Scientific Domain

CI = f(Data, Models, Intelligence, Decisions)

Construction Intelligence is the scientific domain that transforms construction data, practices and documents into measurable models, decision logic and performance-improving outputs. It is not limited to artificial intelligence or software dashboards. It is the discipline of connecting construction facts to decision-making in a structured and auditable way.

6. CI-DF Framework

CI-DF = f(Theory, Science, Framework, System, Platform, Center)

The Construction Intelligence Decision Framework is the backbone linking theory to science, science to models, models to the operational system, the system to the digital platform, and the platform to a real-world application and validation center.

Figure 1. CI-DF Integrated Architecture

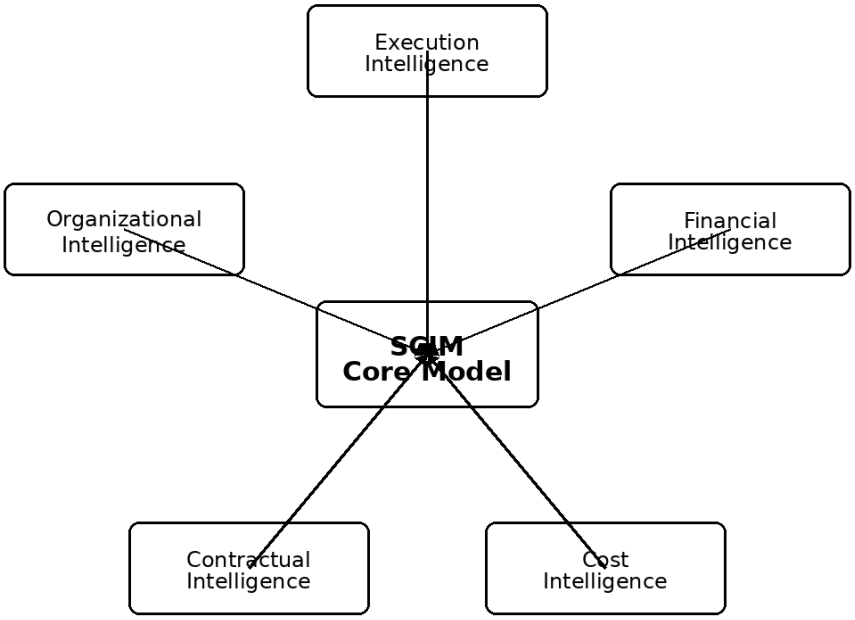
Project outcomes feed updated rules, coefficients and decision thresholds.

7. SCIM: System Construction Intelligence Model

$$SCIM = f(EI, FI, CIc, DI, OI)$$

SCIM is the analytical model that integrates the core dimensions of contracting intelligence. It supports a holistic view in which technical execution, financial viability, cost structure, contractual rights and organizational readiness are evaluated together.

Figure 3. SCIM Intelligence Dimensions



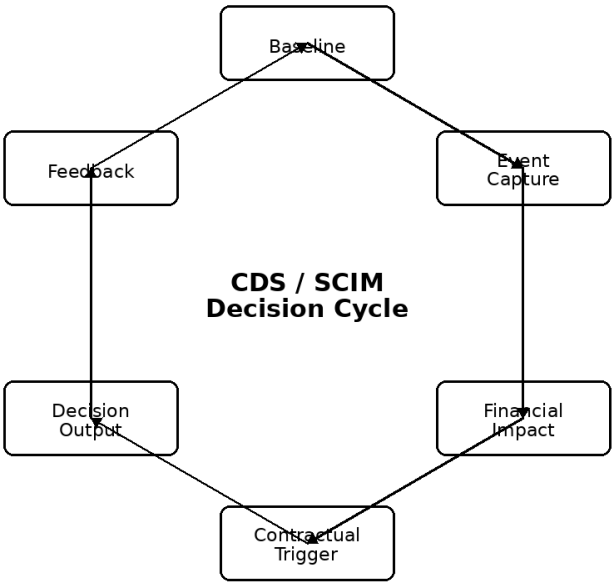
8. CDS: Construction Decision System

CDS = Implementation(SCIM)

Input -> Processing -> Decision -> Output -> Feedback

The Construction Decision System is the operational engine that converts SCIM analysis into decision outputs. Each decision follows an input-processing-decision-output-feedback logic.

Figure 8. Integrated Decision Cycle



9. SITCO Digital Platform and CI Center

Platform = f(Language, Currency, Market, Prices, Engines, Reports)

Center = f(Projects, Training, Validation, Feedback, Development)

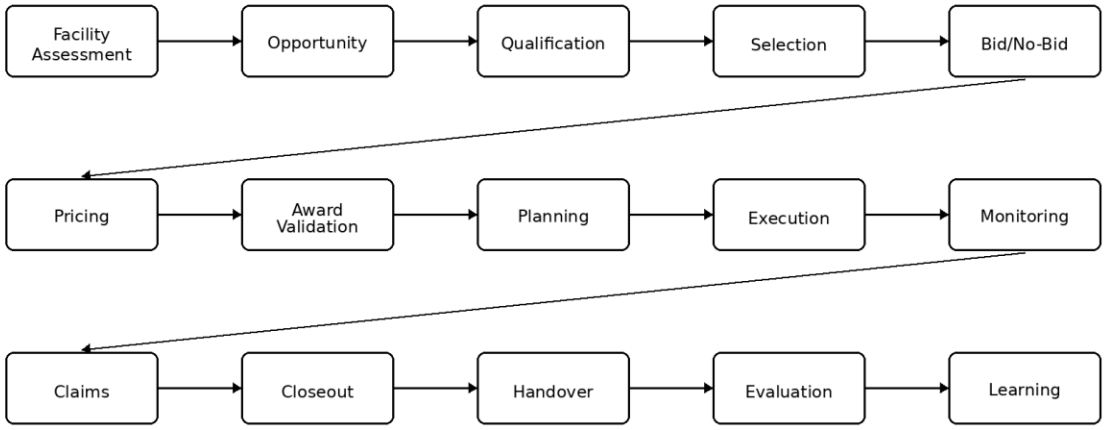
The SITCO Platform is the digital expression of the CI-DF methodology. It is not a visual dashboard only; it is the operating environment where language, currency, market data, prices, engines, reports and decision rules are activated. The CI Center is the application, training, validation and development hub.

10. PLDS: Project Lifecycle Decision System

PLDS = f(Eligibility, Opportunity, Selection, Bid Decision, Pricing, Award, Planning, Execution, Monitoring, Claims, Closeout, Handover, Evaluation, Learning)

PLDS is the lifecycle layer that proves SAITCO does not begin at pricing or execution. It begins before competition, with facility assessment and opportunity screening, and ends after handover, with evaluation and learning.

Figure 2. PLDS Lifecycle Decision Flow



11. GACM and the Spatial-Temporal Applicability Layer

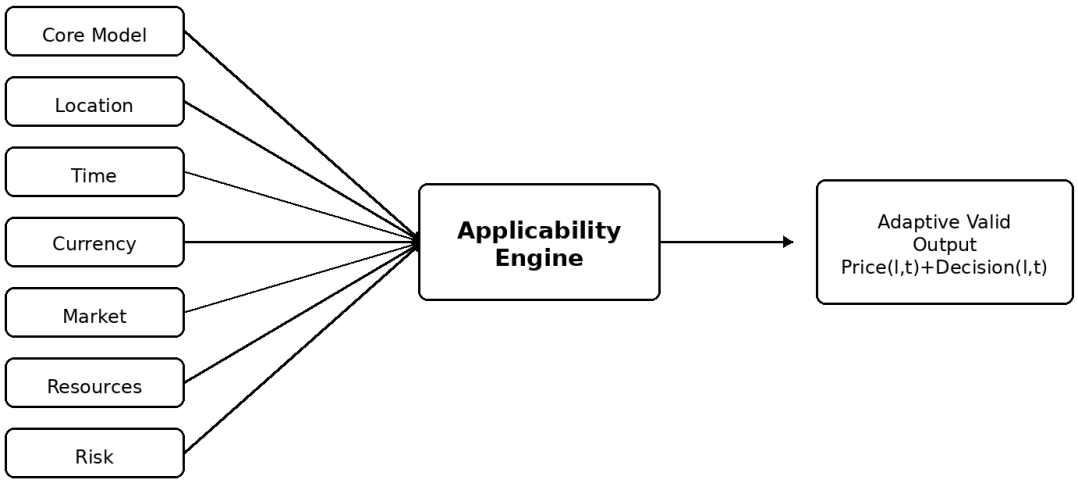
Adaptive Valid Output = f(Core Model, Location, Time, Currency, Market, Resources, Risk)

Price(l,t) = f(Labor(l,t), Materials(l,t), Equipment(l,t), Overheads(l,t), Currency(t), Market(t), Risk(l,t))

The global validity of SAITCO does not mean that the same price or decision should be produced in every country, city, year or market condition. Validity means that the same scientific method can produce different but contextually correct outputs according to local and temporal variables.

The same project in Riyadh, Jazan, Dubai or Cairo is not the same project economically. The same project in 2020 and 2026 is not the same price environment. A model that ignores place, time and market cannot be scientifically valid for construction decision-making.

Figure 4. Spatial-Temporal Applicability Layer



Validity = contextually correct outputs, not identical outputs.

12. SECIM: Self-Evolving Construction Intelligence Model

$$Model(t+1) = Model(t) + Learning(Data(t), Feedback(t))$$

The system does not end when it produces a decision. Each decision creates performance evidence. Each project produces actual productivity, price deviation, claim result, contractual lesson, payment behavior and organizational feedback. SECIM converts this feedback into model development.

13. DCIM and CPOM

$$Cost = f(Labor, Equipment, Materials, Overheads, Risk)$$

$$Performance = f(Time, Cost, Quality, Risk, Efficiency)$$

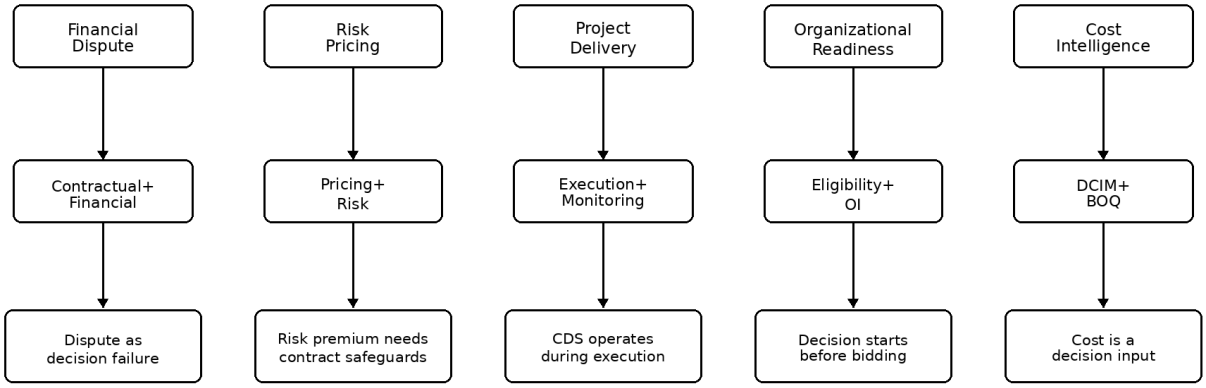
Dynamic Cost Intelligence Model is a layer within the wider decision system. Cost is not the final purpose of SAITCO; it is a decision input. Construction Performance Optimization Model then measures whether decisions improve time, cost, quality, risk and efficiency.

14. Methodology

This study adopts a conceptual system design and multi-case analytical validation methodology. The paper is intended as a first-stage SSRN preprint documenting the intellectual architecture of CI-DF and preparing the foundation for later journal-level empirical validation.

15. Multi-Case Validation

Figure 5. Multi-Case Validation Logic

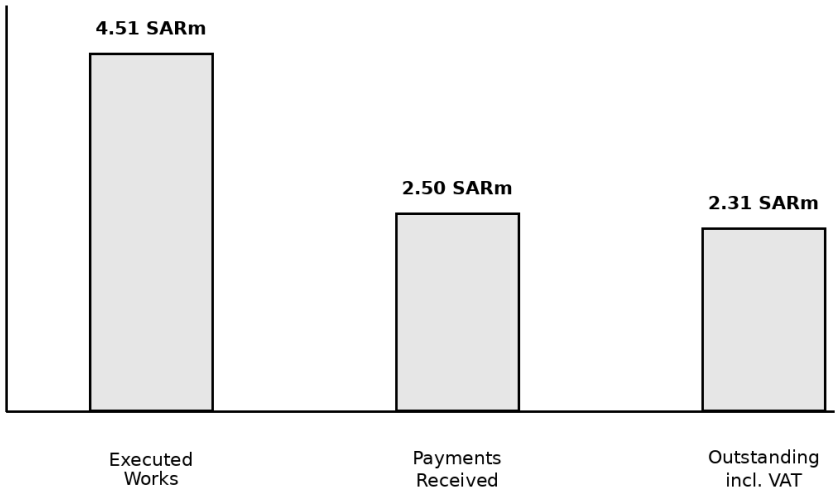


Case	Evidence base	CI-DF layer tested	Main analytical role
Case 1: Financial claim and dispute	Executed value SAR 4,511,960; payments SAR 2,500,567; outstanding SAR 2,313,102 incl. VAT.	Contractual Intelligence + Financial Recovery	Tests the proposition that disputes are delayed decision failures.
Case 2: Risk-adjusted pricing	Sewer infrastructure proposal SAR 13,031,720 excluding VAT; BOQ; 10% risk premium; contractual safeguards.	Pricing + Risk + Contractual Safeguards	Tests the need to protect risk pricing through clauses.
Case 3: Project management / portfolio	212+ projects; 35+ years; SAR 2.06 billion; 120+ claims; 300+ contracts.	Execution + Financial + Contractual Intelligence	Tests integration of project delivery, cost control and claims readiness.
Case 4: Establishment and restructuring	Company establishment, restructuring, policies, qualification and operational efficiency.	Organizational Intelligence + Eligibility	Tests that project decisions begin before bidding.
Case 5: Cost intelligence	BOQ and cost layers; DCIM logic.	Cost Intelligence + BOQ Engine	Tests that cost is a subsystem and decision input.

15.1 Case 1: Financial Claim, Payment Delay and Dispute Escalation

The first applied case concerns a subcontracting financial dispute in a security infrastructure project across two port facilities. The case record reports executed works of SAR 4,511,960 against payments of SAR 2,500,567, leaving an outstanding balance of SAR 2,313,102 inclusive of VAT. This supports the proposition that a dispute is often the visible end of a decision chain: payment delay leads to cash-flow disruption, supplier pressure, weakened negotiation position, settlement pressure and residual dispute.

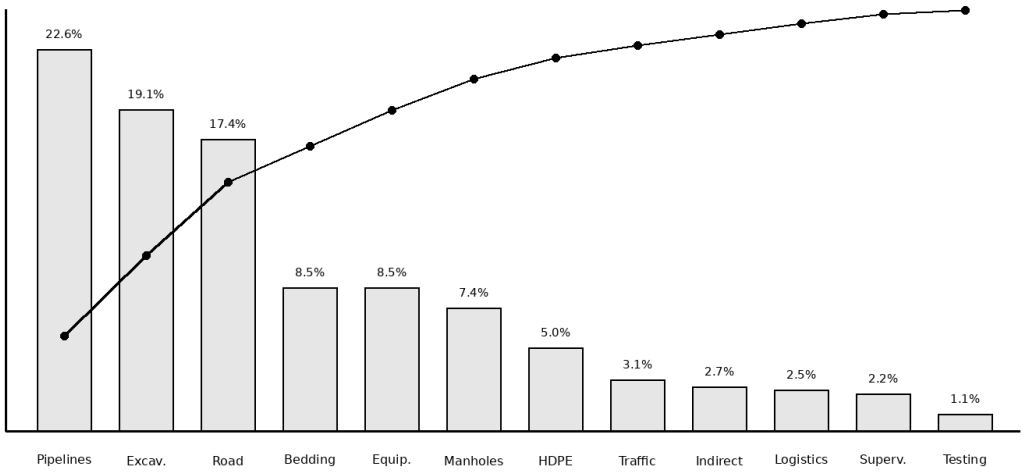
Figure 6. Case 1 Financial Gap Model



15.2 Case 2: Risk-Adjusted Pricing and Contractual Safeguards

The second case concerns a sewer network project in Saudi Arabia with a submitted proposal value of SAR 13,031,720 excluding VAT. The case develops a Pre-Absorbed Risk Pricing and Contractual Control Model (PARP-CCM), where a risk premium is priced into the offer and protected through clauses relating to payment within thirty days, suspension rights, remeasurement, financial rebalancing, time extension and prior approval for additional works.

Figure 7. Case 2 BOQ Pareto Concentration



15.3 Case 3: Project Delivery, Claims Recovery and Portfolio Intelligence

The third case transforms a professional construction portfolio into an applied construction intelligence case. The portfolio evidence includes more than 212 projects, more than 35 years of experience, six principal sectors, an

aggregated executed/studied value of approximately SAR 2.06 billion, more than 120 managed claims, more than 300 analyzed contracts and a reported claim recovery success rate exceeding 80%.

15.4 Case 4: Organizational Structuring and Qualification Readiness

The fourth case introduces organizational intelligence as a necessary layer of project performance. The SAITCO-OI model is structured into company establishment engineering, administrative and financial restructuring, policy and procedure development, qualification and classification readiness, and operational efficiency optimization. The case evidence indicates more than five companies established or restructured and more than fifteen restructuring and qualification projects executed.

$$OI = f(CE, RS, PS, QC, OE)$$

15.5 Case 5: Cost Intelligence and BOQ-Based Decision Input

The fifth case validates DCIM by treating cost analysis as a subsystem within the wider decision framework. In conventional practice, cost may be treated as the final output of estimating. In CI-DF, cost is a dynamic input into eligibility, opportunity selection, bid strategy, risk pricing, payment planning, claims valuation and closeout.

16. Cross-Case Analysis

Lifecycle stage	Dominant failure type	Observed outcome	CI-DF corrective mechanism
Before competition	Weak qualification and organizational readiness	Entry into unsuitable projects	Facility assessment + OI + PLDS selection logic
Pricing	Risk ignored or not contractually protected	Financial loss and margin erosion	DCIM + PARP-CCM + contractual safeguards
Contracting	Weak protection, unclear final account or payment rules	Disputes and residual claims	DI + notice matrix + valuation logic
Execution	Poor integration of site, finance and contract data	Deviation, evidence loss and cost leakage	SCIM + CDS live decision cycle
Monitoring	Late detection of claim triggers	Escalation and weakened bargaining position	Early warning + evidence preservation + claim readiness
Closeout	Incomplete reconciliation	Settlement uncertainty and residual claims	Final account validation + expert determination logic

Across the cases, the same pattern repeats. Damage does not appear suddenly. It matures through decisions that were missing, late, poorly documented, underpriced, unprotected or disconnected from other layers of the project system. The cross-case conclusion is therefore: construction disputes are delayed system failures.

17. Discussion

The discussion can be summarized in five paradigm shifts. First, a construction project is not merely execution; it is an integrated decision system. Second, a dispute is not merely a legal event; it is often a delayed manifestation of previous decision failure. Third, pricing is not arithmetic only; it is a risk-adjusted decision connected to contract protection, cash flow and market behavior. Fourth, management is not only monitoring; it is continuous decision control. Fifth, accuracy alone is not enough; applicability defines correctness.

The Spatial-Temporal Applicability Layer is especially important. A model can be mathematically accurate and practically wrong if it ignores local labor productivity, material availability, equipment cost, currency, taxes, supply chains, approvals, climate, market volatility and project timing. For construction intelligence, a valid output is not the same output everywhere; it is the output that is correct for the specific environment.

18. Integrated System Definition

$$SAITCO\text{-}CI = f(Problem, Theory, CI, CI\text{-}DF, SCIM, CDS, Platform, Center, Lifecycle, Global\ Adaptation, Self\text{-}Learning)$$

SAITCO Construction Intelligence System = f(CI, CI-DF, SCIM, CDS, Platform, Center, Lifecycle, Global, Learning)

SAITCO / CI-DF is a full lifecycle construction decision intelligence system. It is not a cost model alone, not a project management tool alone, not a dashboard alone, and not a legal claims template alone. It is an integrated decision-driven ecosystem governing the construction process from pre-bid eligibility to post-project evaluation and continuous learning.

19. Implementation Roadmap and Publication Outputs

The immediate output of this package is a SSRN-ready preprint, cover letter, metadata file and upload checklist. Future outputs may include a SAITCO white paper, copyright deposit, figure package, software requirements document and CI Center business plan.

20. Limitations and Future Research

This manuscript is a first-stage conceptual and applied preprint. It establishes the framework and validates its logic through five applied cases. It does not yet claim statistical generalization from a randomized dataset. Future research should test CI-DF using independent multi-project datasets, quantitative performance measures, before-and-after organizational implementation and external expert review of claim outcomes.

21. Conclusion

This paper presents Construction Intelligence as a decision science and introduces SAITCO / CI-DF as an integrated framework for transforming engineering project contracting into an adaptive, lifecycle-based, self-learning digital intelligence system. The research shows that the construction sector's persistent failures are not only technical or legal problems; they are decision architecture problems.

The five applied cases demonstrate that financial disputes, risk pricing, project delivery, organizational readiness and cost analysis are not separate topics. They are different expressions of the same system problem: fragmented decisions. CI-DF addresses this by making every stage of the project lifecycle a decision node and every project outcome a source of learning. The core conclusion is that construction disputes are not merely legal events; they are delayed system failures.

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Appendix A. SSRN Upload Metadata

Field	Recommended entry
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